

HASSAN FAISAL

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Professional Summary

Computer Science graduate with practical experience developing intelligent software solutions, AI-powered applications, and automated processing systems. Proficient in Python and modern AI technologies, with hands-on experience in Generative AI, Large Language Models, Retrieval-Augmented Generation (RAG), Agentic AI, LangGraph, LangChain, and vector databases. Experienced in designing and implementing end-to-end solutions, integrating AI capabilities into software applications, and developing efficient automation workflows. Strong problem-solving mindset with a focus on building practical, scalable, and reliable solutions for real-world applications.

Education

Bachelor of Science in Computer Science

03/2022 – 01/2026

National College of Business Administration & Economics • CGPA: 3.50/4.0

- Completed a Bachelor of Science in Computer Science with a strong focus on practical software development and AI application development.
- Co-authored two conference papers in Machine Learning, gaining experience in literature review, experimentation, and research evaluation methodologies.
- Explored emerging AI technologies through academic and independent projects, including Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), and Agentic AI systems.

Projects

Multi-Agent Customer Support System

Multi-Agent AI Systems • LangGraph • Retrieval-Augmented Generation (RAG) • OpenAI API • Vector Search

- Architected a multi-agent customer support platform using LangGraph and FastAPI, coordinating specialized agents for query classification, knowledge retrieval, response generation, and fallback handling.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline with Pinecone vector search and OpenAI models to generate context-aware, source-grounded responses from enterprise knowledge bases.
- Benchmarked system quality using the RAGAS evaluation framework, iteratively optimizing prompt design, retrieval strategies, and agent workflows to maximize faithfulness, answer relevancy, and context precision.

Automated Photogrammetric Processing Framework

Photogrammetry Pipeline Automation • Python • OpenDroneMap (ODM) • WebODM • Docker

- Engineered an automated photogrammetric processing framework leveraging OpenDroneMap (ODM) to generate high-quality Orthomosaics, DSMs, and DTMs from aerial imagery through reproducible processing pipelines.
- Benchmarked multiple ODM execution profiles by analyzing reconstruction accuracy, execution time, and hardware utilization to establish optimized configurations for diverse operational requirements.
- Developed automation workflows for dataset preprocessing, task execution, and output management, improving processing efficiency and minimizing manual overhead.

Healthcare Appointment Coordination System

Healthcare Appointment Management • Python • FastAPI • PostgreSQL • ReactJS

- Full-stack healthcare appointment scheduling platform built using **FastAPI**, **ReactJS**, and **PostgreSQL**, enabling patients to book appointments and doctors to manage availability.
- Developed secure RESTful APIs with **JWT authentication** and **role-based access control (RBAC)** for patients, doctors, and administrators.
- Designed an administrative dashboard for managing doctor schedules, appointments, and platform operations, improving healthcare workflow efficiency.

Professional Experience

Artificial Intelligence Apprentice

03/2026 – Present

Deep Cognitive Solutions, Lahore, Pakistan

- Contributed to the development of AI-powered applications by supporting machine learning workflows, experimentation, and software development across production-oriented projects.
- Participated in the evaluation and optimization of large-scale image processing pipelines through systematic experimentation and performance analysis.

- Developed Python-based automation scripts to streamline repetitive processing tasks and improve workflow efficiency.
- Explored modern AI paradigms including Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Agentic AI for building intelligent and autonomous systems.
- Collaborated with AI engineers in the design, implementation, and iterative improvement of AI-driven software solutions.

Awards & Achievements

Best Product Award & 2nd Place Overall

July 2026 – August 2026

Comeback Pakistan

- Selected among **65 participants from 1,000+ applicants** and secured **2nd place overall** in the program.
- Served as **Technical Lead**, owning the technical lifecycle from **R&D and solution development to deployment**, while working with the team to validate the product and take it toward **potential business adoption**.
- Drove technical execution, product iteration, and deployment while collaborating with the team to position the solution for **real business use cases**, ultimately earning the program's **Best Product Award**.

Technical Skills

Programming: Python

Generative AI: OpenAI API, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Agentic AI, Prompt Engineering

AI Frameworks & Automation: Model Context Protocol (MCP), n8n, AI Workflow Automation, RAGAS

Backend: FastAPI, API Integration

Vector Databases: Pinecone, Chroma, FAISS

Tools: Docker, Git, GitHub, Ollama, Postman, Jupyter Notebook

Selected Certifications & Training

[Claude Code in Action](#)

Anthropic